

Daily Quiz - 4 (ELL782/ELL7282) - Computer Architectures

Instructions

- **Time:** 5 minutes
- Maximum marks: 10
- Number of questions: 10
- Each question carries **1 mark**.
- Questions marked **MCQ** have exactly one correct option.
- Questions marked **MSQ** may have one or more correct options. For an MSQ, select all correct options and no incorrect option.
- There is **no partial credit** for an MSQ.

Points: 10/10 Time: 01:58

Name * 

test

Entry Number * 

test

✓ **Correct** 1/1 Points

The 32-bit value 0x87654321 is stored beginning at byte address 0x100 using little-endian representation. What bytes are stored at addresses 0x101 and 0x103, respectively? 

☐ (0x87, 0x21)

☒ (0x43, 0x87)

☐ (0x21, 0x65)

☐ (0x65, 0x43)

✓ **Correct** 1/1 Points

An array `int a[3][4]` begins at address `0x200`; each integer occupies 4 bytes. What is the address of `a[1][2]` under row-major and column-major storage, respectively? 

☐ (0x21C, 0x21C)

☐ (0x218, 0x218)

☒ (0x218, 0x21C)

☐ (0x21C, 0x218)

✓ **Correct** 1/1 Points

Consider the assembly statement "**loop: add r1, r2, 10 @ update total.**" Which statements about it are correct? **Select all that apply.** 

☒ `add r1, r2, 10` is in 3-address format.

☒ `loop` is a label.

☒ `10` is an immediate operand.

☐ `@ update total` changes the machine-level result of the instruction.

✓ **Correct** 1/1 Points

Which mapping of GNU assembly-file components to their meanings matches the lecture slides? 

☐ `.file` – instructions; `.text` – source-file name; `.data` – constants

- ☐ .file – source-file name; .text – constants; .data – instructions
- ☐ .file – constants; .text – instructions; .data – source-file name
- ☒ .file – source-file name; .text – instructions; .data – constants/read-only variables

✓ **Correct** 1/1 Points

Suppose $r1 = 1000$, $r2 = 24$, and $pc = 4000$. Which operand uses base-index-offset addressing and accesses memory address 1124? 

- ☐ [1124]
- ☐ 100[r1]
- ☒ 100[r1,r2]
- ☐ 100[pc]

✓ **Correct** 1/1 Points

SimpleRisc immediate operands are 16-bit two's-complement values. Which pair can both be used as immediate operands in mov? 

- ☐ -32769 and 32767
- ☒ -32768 and 32767
- ☐ -32768 and 32768
- ☐ -65535 and 65535

✓ **Correct** 1/1 Points

After executing the following SimpleRisc code, what value is stored in r4?

```
mov r0, 29
mov r1, 6
```

```
div r2, r0, r1  
mod r3, r0, r1  
sub r4, r3, r2
```

☐ 5☐ 23☒ 1☐ 4

✓ **Correct** 1/1 Points

Before executing `cmp r1, r2`, both `r1` and `r2` contain 12. Which statements are correct after the instruction? **Select all that apply.**

☐ `r1` is overwritten with 0.☒ The result of the comparison is recorded in the special flags register.☒ `flags.E = 1`☒ `cmp` is one of SimpleRisc's six arithmetic instructions.

✓ **Correct** 1/1 Points

The program counter contains 0x1000. In the PC-relative operand 100[pc], the offset 100 is decimal. What effective memory address is accessed?

☒ 0x1064☐ 0x0F9C☐ 0x1100☐ 0x10064

✓ **Correct** 1/1 Points

A row-major array `int a[3][4]` begins at address `0x200`. Register `r1` contains `0x200`. Which base-offset operand accesses `a[2][1]`? Assume each `int` occupies 4 bytes. 

☐ `28[r1]`

☒ `36[r1]`

☐ `24[r1]`

☐ `40[r1]`

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